AIU. ALAMEIN INTERNATIONAL UNIVERSITY
جامعة العلمين الدولية

Smart Surveillance System to Detect Unusual Activities in Real Time

CSE/AIE Graduation Project Presentation

Supervised by : Dr Essam Abdellatef

15/6/2026



AIU. ALAMEIN INTERNATIONAL UNIVERSITY جامعة العلمين الدولية

Team Members



Ahmed Mostafa
Software Engineer



Mohamed Ahmed
Software Engineer
Team Lead



Abdullah Abdalgawad
AI Engineer



Khalid Abdelsalam
Software Engineer



Omnia Adel
AI Engineer



Tasneem Ibrahim
Ai Engineer



Mariam Elrafei
Software Engineer

AIU ALAMEIN INTERNATIONAL UNIVERSITY جامعة العلمين الدولية

SecureSight — Real-Time Intelligent Surveillance System for Video Anomaly Detection



1. Massive CCTV footage is under-analyzed

Vast amounts of recorded video data remain unreviewed in real time, leading to potential oversights.



2. Human monitoring suffers from attention fatigue

Operators cannot maintain high vigilance for extended periods, increasing the risk of missed incidents.



3. Delayed incident detection and response

Critical anomalies are often detected too late, slowing down necessary interventions and reaction times.



4. Lack of smart role-specific emergency alert routing

Many systems fail to automatically direct alerts to the appropriate authorities based on the type of incident.

The challenge is not only detecting anomalies, but detecting them fast, reliably, and sending the alert to the right responder.

MOTIVATION

From passive recording to proactive real-time emergency response.

- Faster detection of dangerous or abnormal activities
- Reduced dependence on continuous human monitoring
- Automatic routing of alerts to police, hospital/EMS, or firefighters
- Better operational reliability through offline alert buffering
- Support safer public spaces and faster emergency response



Police

Hospital/EMS

Firefighters

AIU

Literature Review

Research Background & The SecureSight Gap



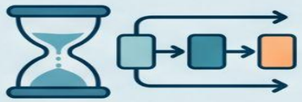
1. Reconstruction-Based Methods

- **Autoencoders** detect anomalies using reconstruction error.
- ⚠ **Limitation**: Anomalies may still be reconstructed well.



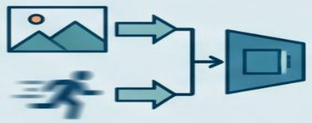
2. Weakly Supervised MIL Methods

- **UCF-Crime** introduced video-level anomaly learning.
- ⚠ **Limitation:** Coarse labels and limited precise localization.



3. Transformer-Based Video Models

- Capture long-range temporal dependencies.
- ⚠ **Limitation:** Often focus on model performance more than deployment.



4. Two-Stream Networks

- Combine spatial appearance and temporal motion.
- ⚠ **Gap:** Many systems lack full real-time alerting and operational resilience.



Research Gap: Need a system that combines spatial + temporal detection, sequence reasoning, **real-time alert routing, and deployment reliability.**

Presenter Alismnn, PMD
Studest SecureSight, PA, LhD
@redrseucsom.org

Recent Published papers limitations

Paper	Data Issues	Bias	Evaluation	Generalization	Model Issues	Temporal Issues	Training Issues	System / Deployment	Efficiency	Other
Shoplifting (YOLOv5)	Dataset, Representativeness, Imbalance, Truncation	Bias	Metrics	Generalization	Features, Robustness, Tracking	—	—	Deployment	—	—
Survey (Suspicious Behavior)	—	Bias	Metrics, Evaluation	—	Modality	—	—	Deployment	—	Selection, Coverage, Rigor, Definition, Ethics, Clarity
Spatiotemporal Autoencoder	Small Datasets, Low Diversity	Reconstruction Bias	—	Poor Generalization	—	—	Manual Thresholds	—	—	Fixed Cameras, No Ablation
Wu et al. (2024)	—	Benchmark Bias	Weak Analysis	—	—	—	—	—	—	Inconsistent Comparisons, No Meta-analysis, Limited Coverage
Song (2022)	—	Selection Bias	Weak Evaluation, No Metrics	—	—	—	—	—	—	Shallow Review, No Failure Analysis
Real-world AD	Limited Data, Class Imbalance	—	—	Weak Generalization	—	—	Sparse Training	—	—	No Fairness
RTFM	—	—	—	No Generalization	Backbone Dependency	—	Label Noise	—	—	Fragile Assumptions, Fixed Segmentation
Object-Augmented VAD	—	COCO Bias	—	Weak Generalization	—	Temporal Misalignment	—	—	No FPS	Top-K Limitation
Future Frame Prediction	—	—	PSNR Weakness	—	—	Short Horizon	—	—	—	Low Resolution, No Uncertainty, Flow Assumptions
ViT	Data Hungry	—	—	—	High Complexity	—	—	No Deployment	Scaling Issues	Poor Small Data
Our Paper	—	—	Weak Evaluation	—	—	—	—	Prototype System	Latency Gaps	Unfair Comparison, Non-standard Split

Research Gap / What We Learned

SecureSight — Real-Time Intelligent Surveillance System for Video Anomaly Detection

Based on a literature review and limitations of recent published papers, slide summarize, the key research gaps that motivated our proposed system.

Weak Real-Time Deployment

Many studies focus on model accuracy but do not provide a complete real-time operational alerting system.

Limited Alert Routing

Existing systems often detect anomalies but do not route alerts to the correct emergency authority.

Incomplete Spatial-Temporal Understanding

Some approaches focus mainly on appearance or motion, but not both in a unified practical framework.

Weak Evaluation and Generalization

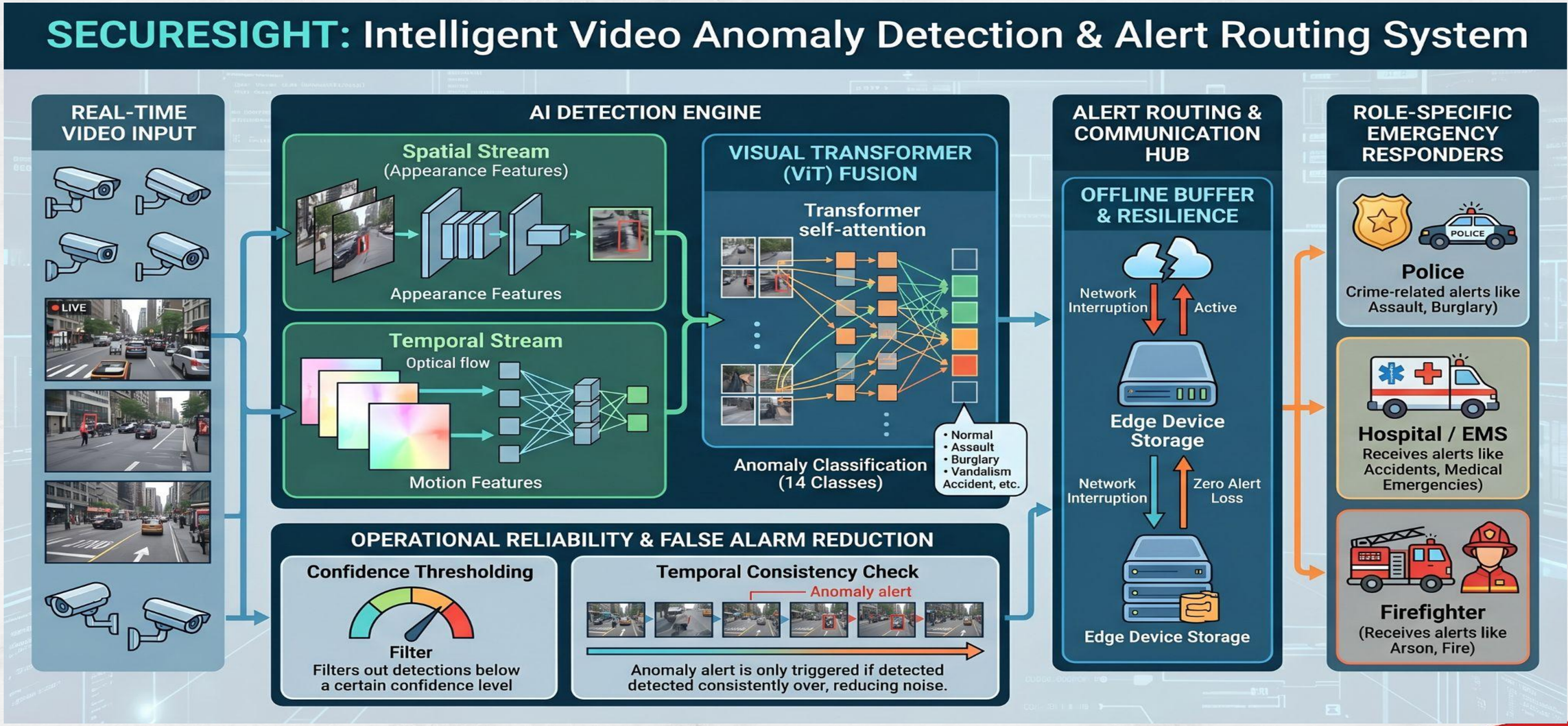
Several papers suffer from weak evaluation protocols, class imbalance, or limited testing under real CCTV conditions.

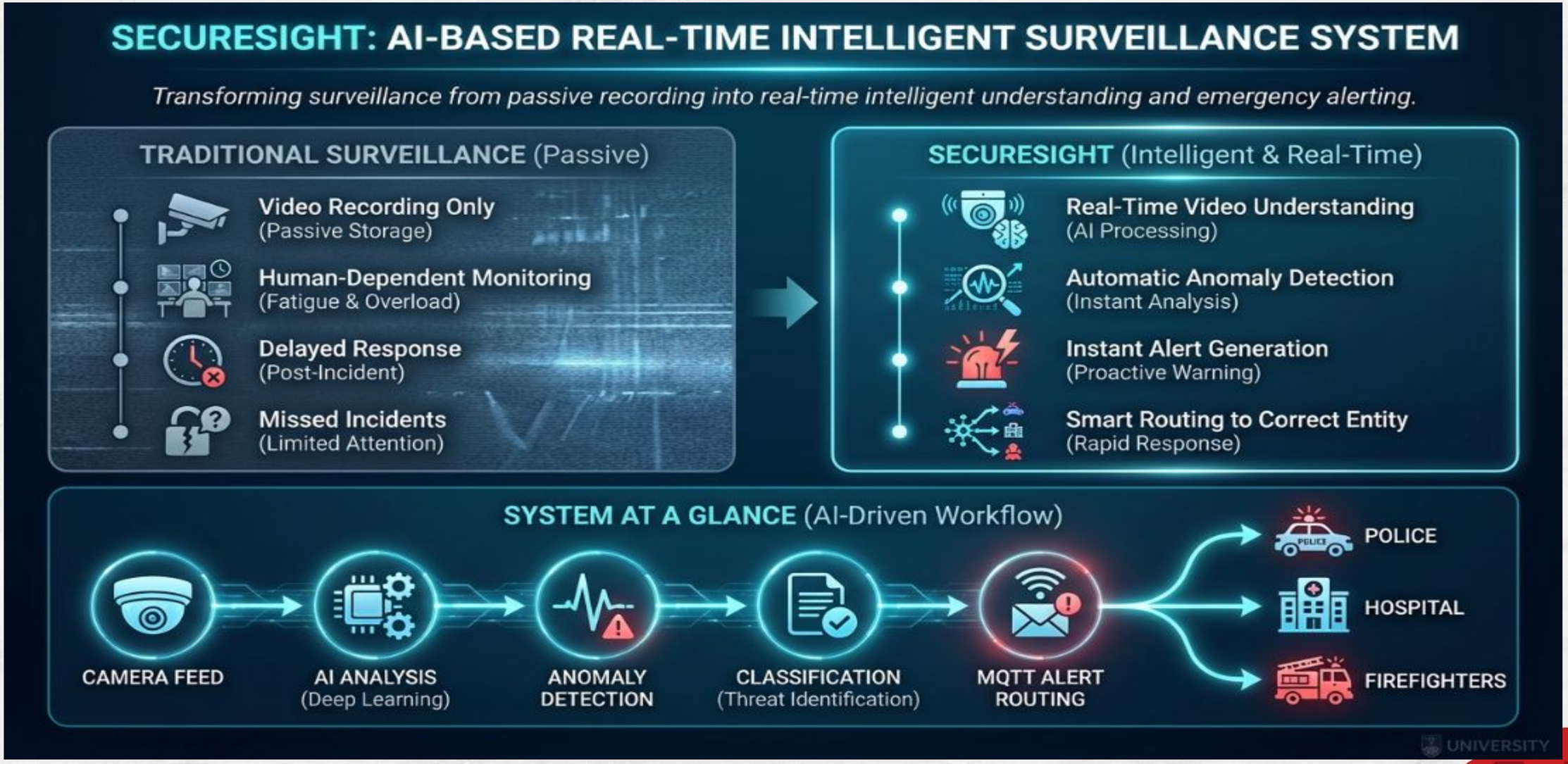
Reliability Gap

Many systems do not address network interruptions, alert continuity, or deployment resilience.

Therefore, **SecureSight was designed as an end-to-end system** combining spatial-temporal anomaly detection, ViT-based temporal reasoning, MQTT alert routing, role-specific dashboards, and offline alert buffering.

Proposed Solution




SECURESIGHT: AI-BASED REAL-TIME INTELLIGENT SURVEILLANCE SYSTEM
Transforming surveillance from passive recording into real-time intelligent understanding and emergency alerting.
TRADITIONAL SURVEILLANCE (Passive)
Video Recording Only
(Passive Storage)
Human-Dependent Monitoring
(Fatigue & Overload)
Delayed Response
(Post-Incident)
Missed Incidents
(Limited Attention)
SECURESIGHT (Intelligent & Real-Time)
Real-Time Video Understanding
(AI Processing)
Automatic Anomaly Detection
(Instant Analysis)
Instant Alert Generation
(Proactive Warning)
Smart Routing to Correct Entity
(Rapid Response)
SYSTEM AT A GLANCE (AI-Driven Workflow)
CAMERA FEED
AI ANALYSIS
(Deep Learning)
ANOMALY
DETECTION
CLASSIFICATION
(Threat Identification)
MQTT ALERT
ROUTING
POLICE
HOSPITAL
FIREFIGHTERS

Requirements

Functional Requirements

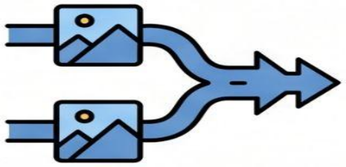
- Accept CCTV/video input
- Extract frames and 16-frame clips
- Preprocess video: resize, normalize, augment
- Classify activity as normal or anomalous
- Generate real-time alerts
- Route alerts to police, hospital/EMS, or firefighters
- Buffer alerts during network interruption

Non-Functional Requirements

- Low-latency inference
- Reliable alert delivery
- Scalable modular architecture
- Role-specific dashboards
- Maintainable system design
- Privacy-aware video handling

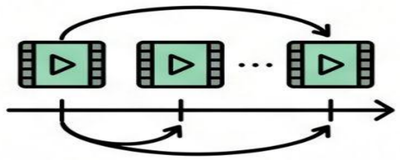
Requirements combine AI accuracy with real-world system reliability.

Why Our Solution Is Strong



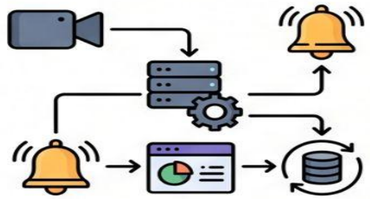
Two-Stream Understanding

Combines spatial appearance and temporal motion.



ViT Temporal Reasoning

Captures long-range dependencies across video clips.



Real-Time Deployment Design

Flask inference, MQTT alert routing, dashboards, and offline buffer.

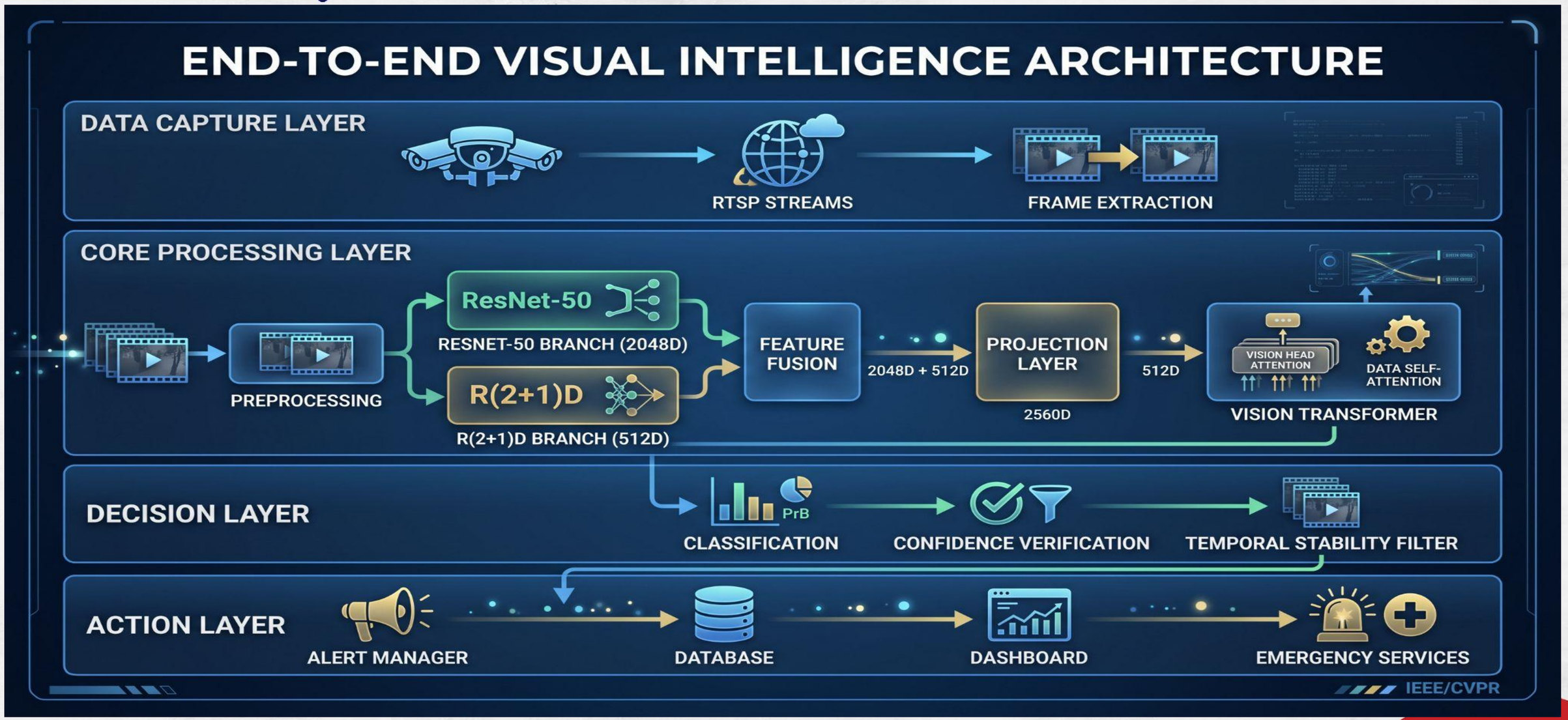


Validated Performance

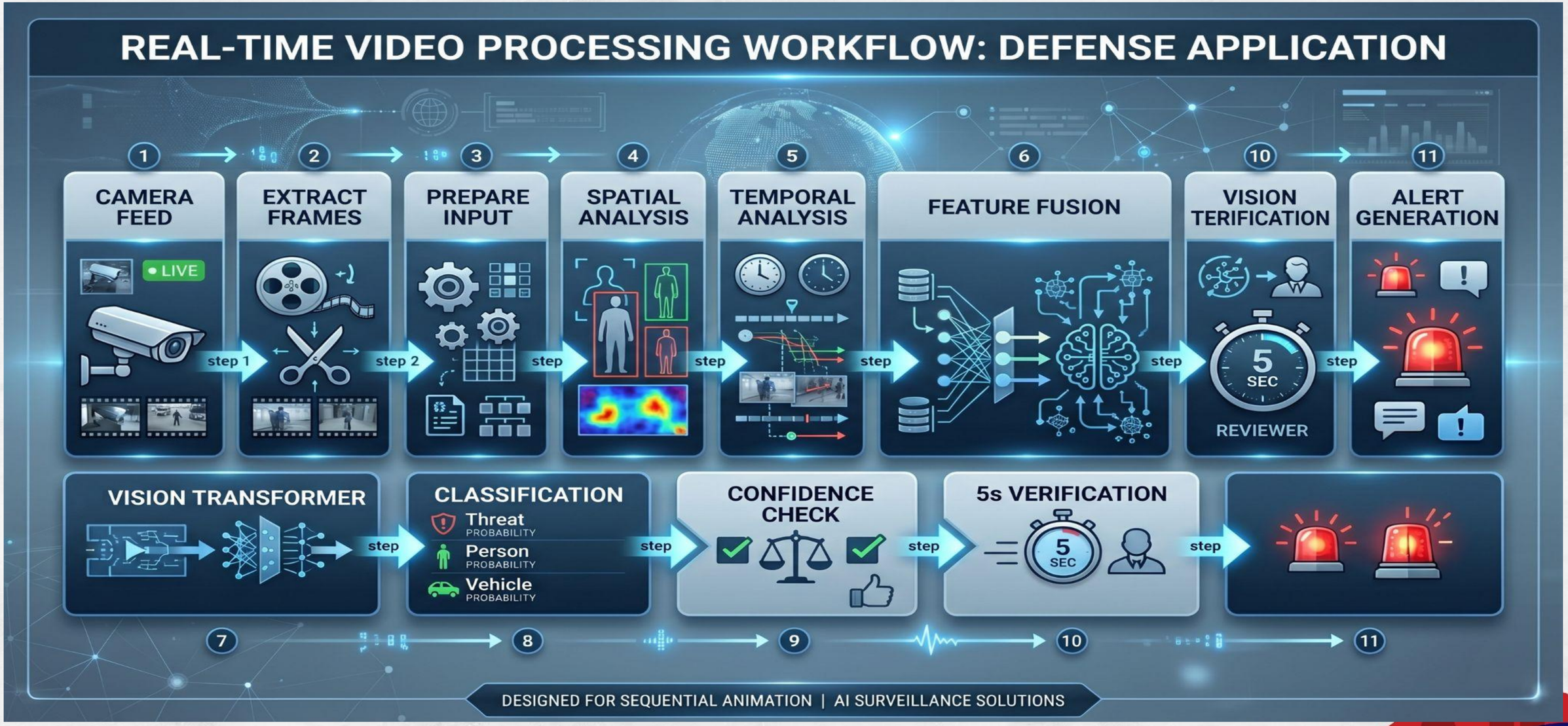
74.3% top-1 accuracy and 0.89 mean AUC on UCF-Crime.

The project is not only a model; it is an end-to-end intelligent surveillance prototype.

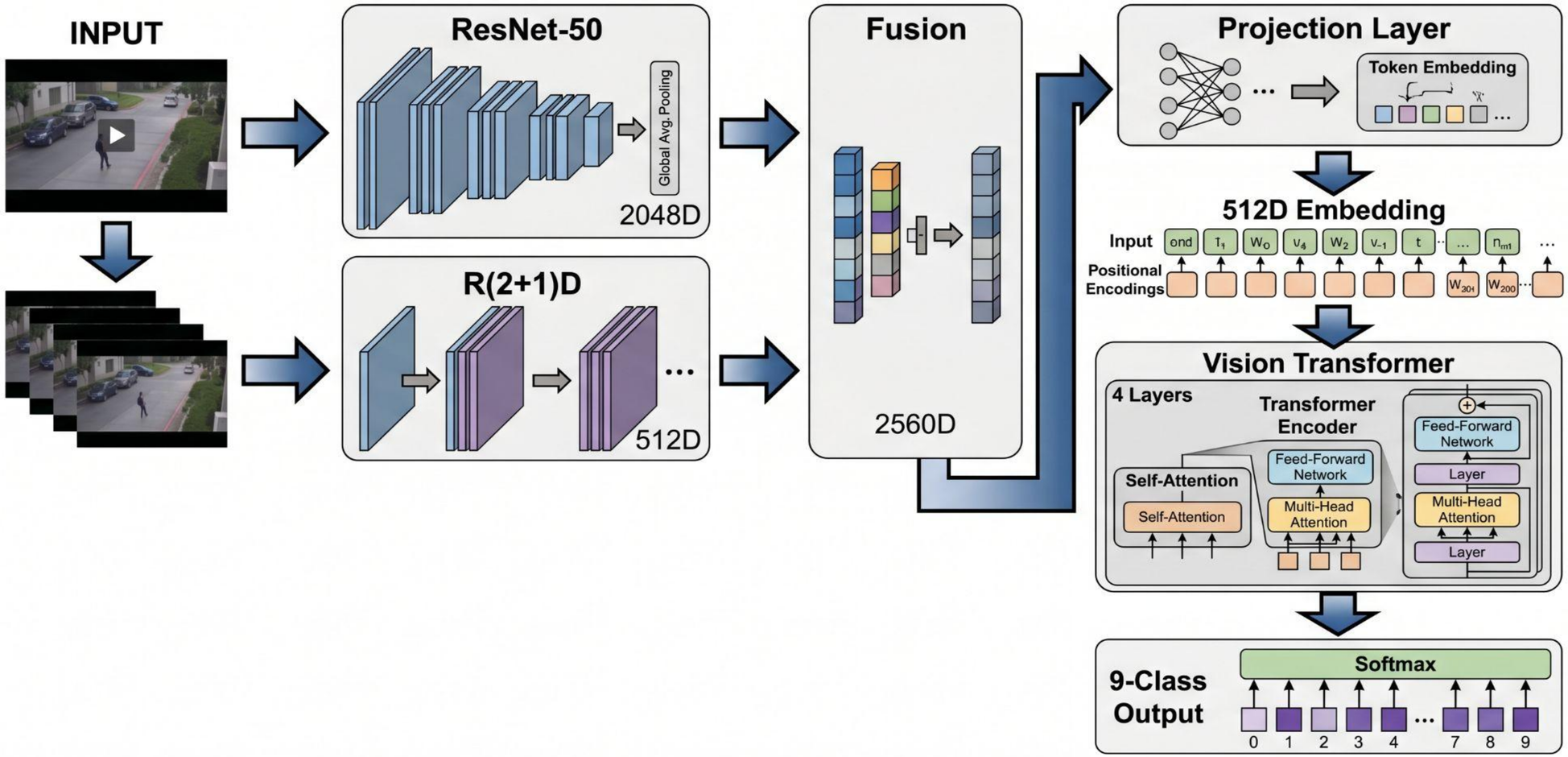
System Architecture Layers



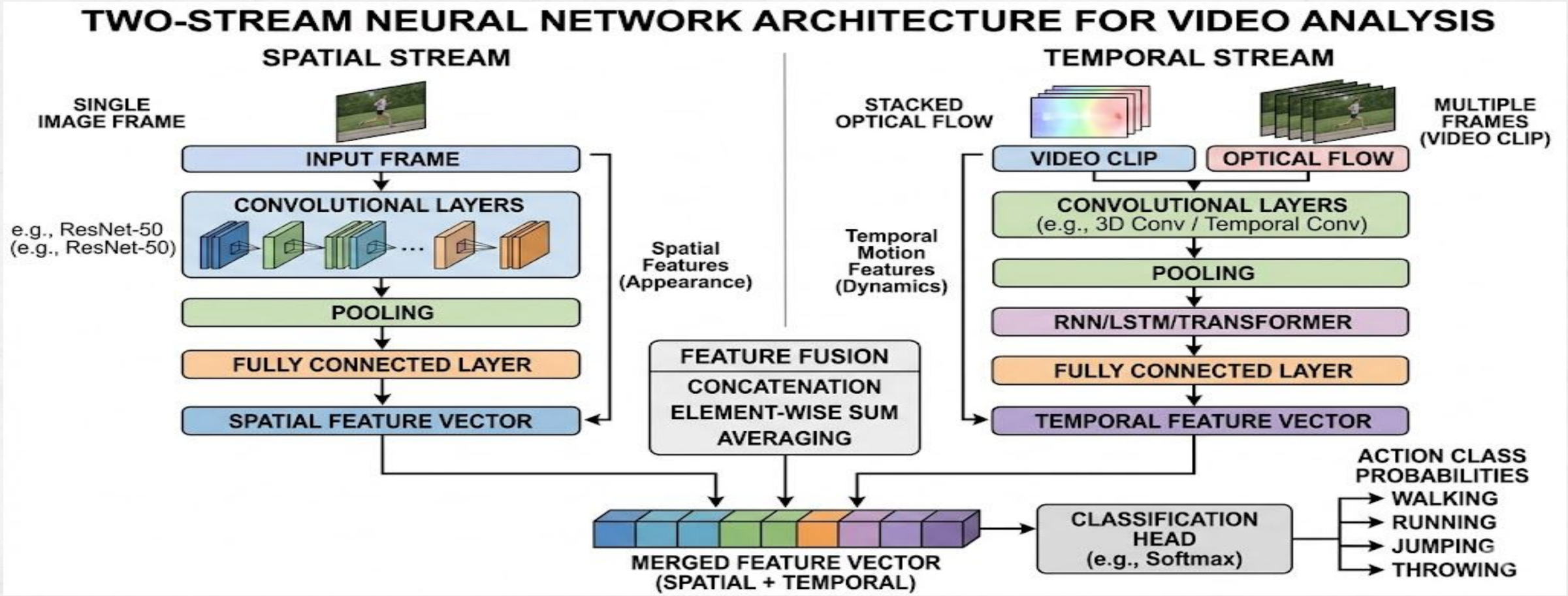
System Workflow



Model Architecture

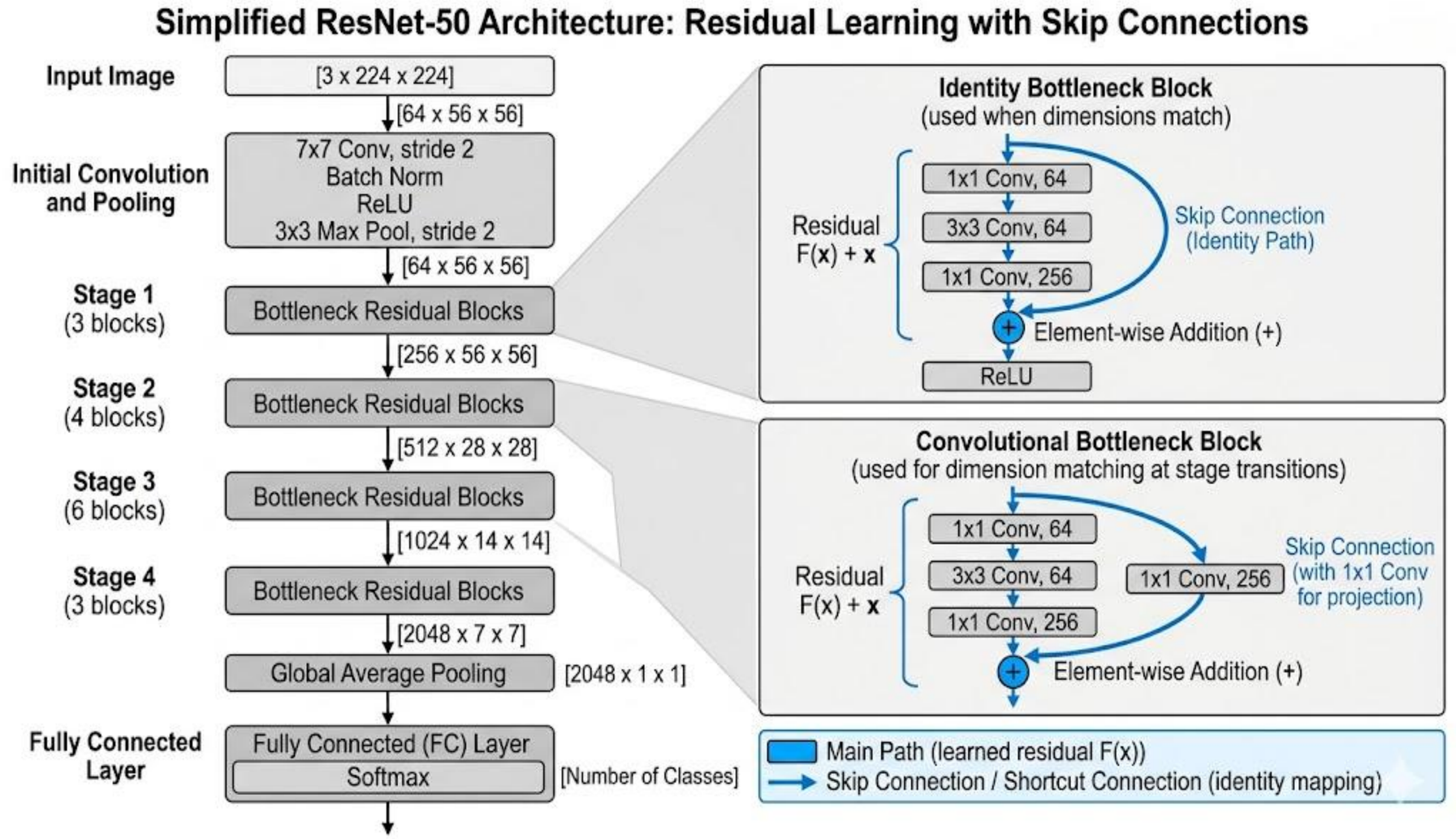


Two-Stream Architecture



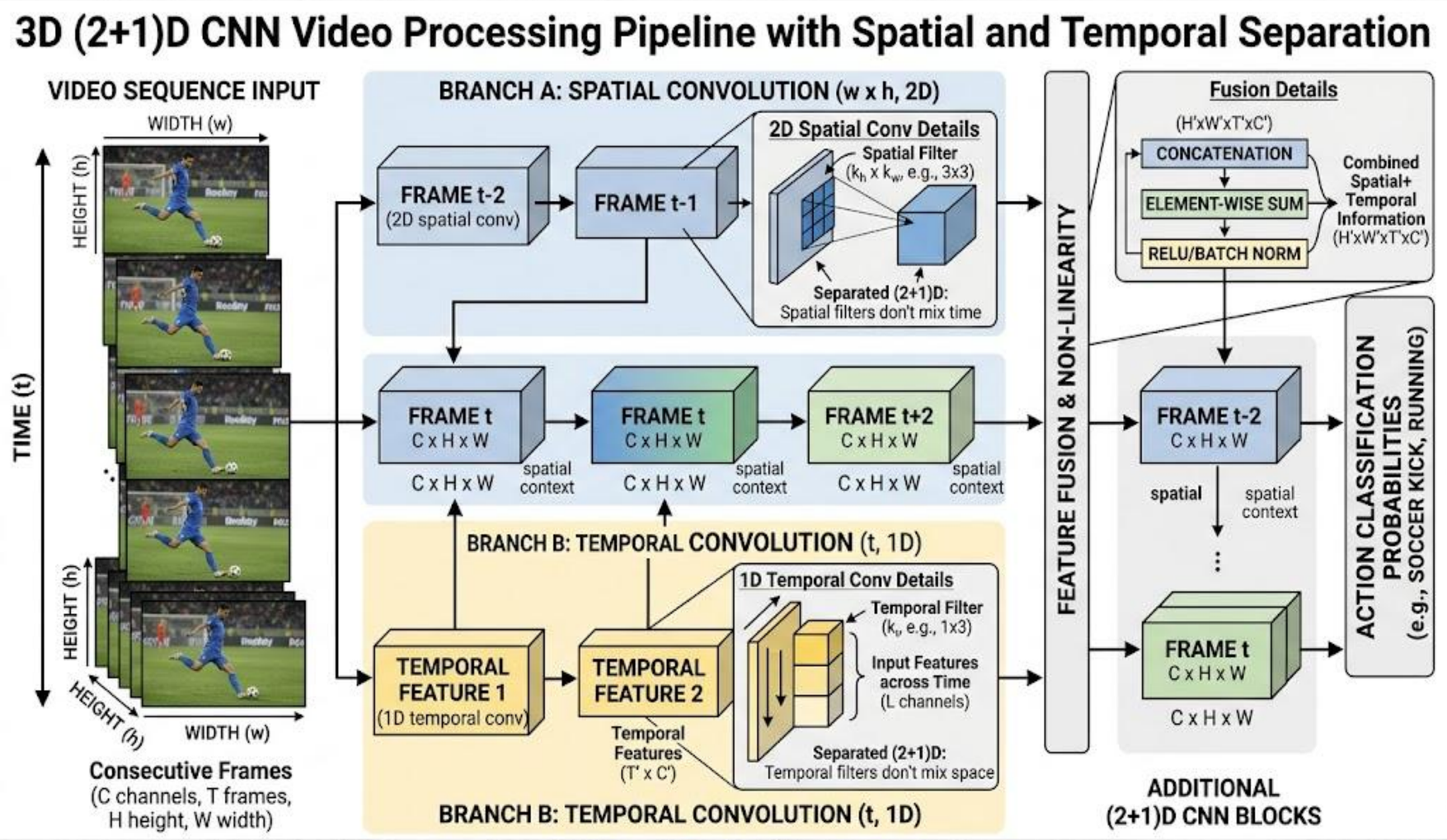
- **Formulation:**
  - ➔ Ffinal=Fuse(Fs,Ft)

Spatial Stream (ResNet-50)



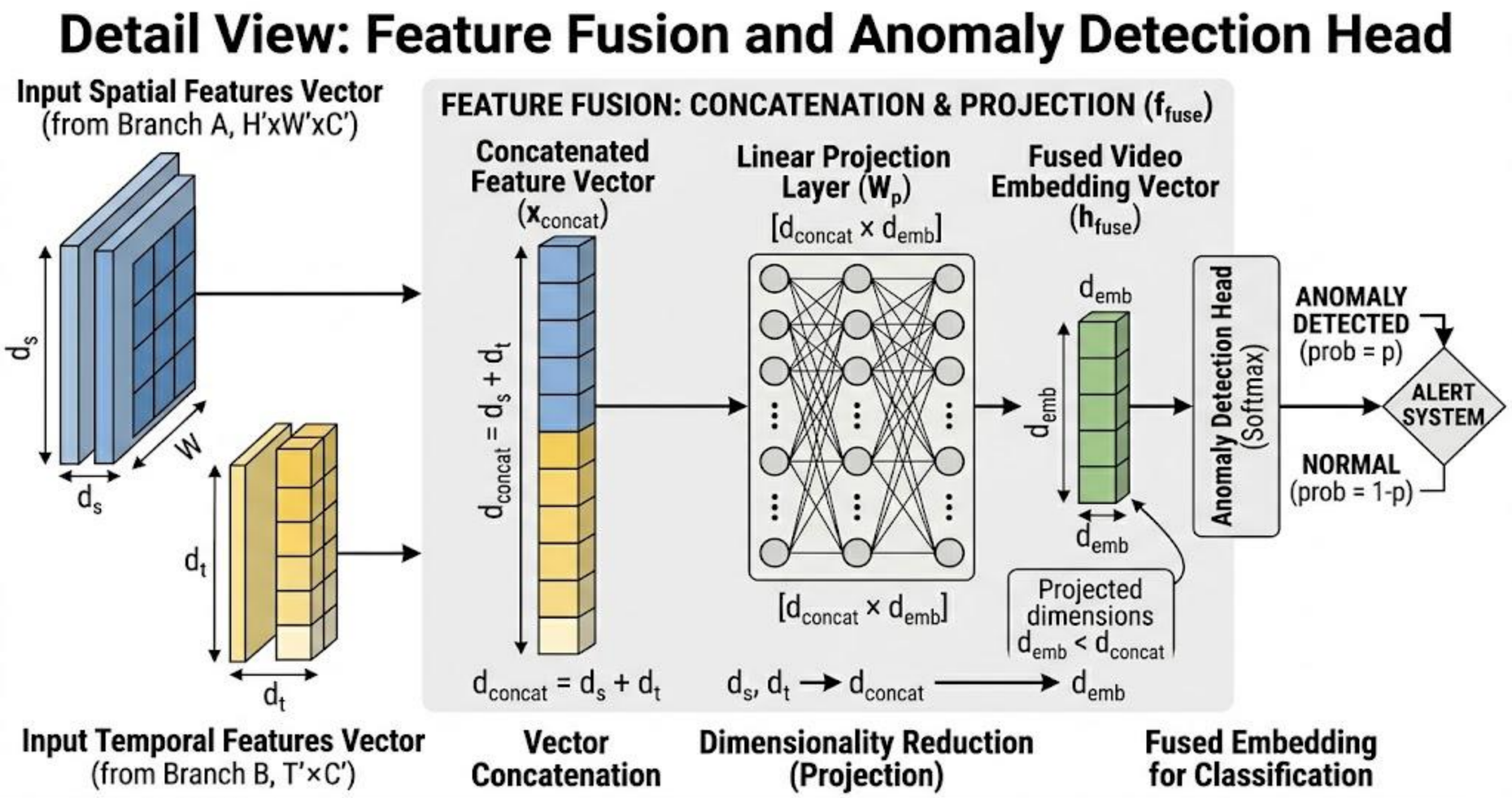
- **Spatial convolution:**
  - ➔ **Fs1 = f (Ws ∗ xt + bs)**
- **Residual Learning:**
  - ➔ **Fs = F (Fs1 + xt)**
- **Final Spatial Output:**
  - ➔ **Ys = f (Ws ∗ Xt + bs) + Xt**

Temporal Stream (R(2+1)D)



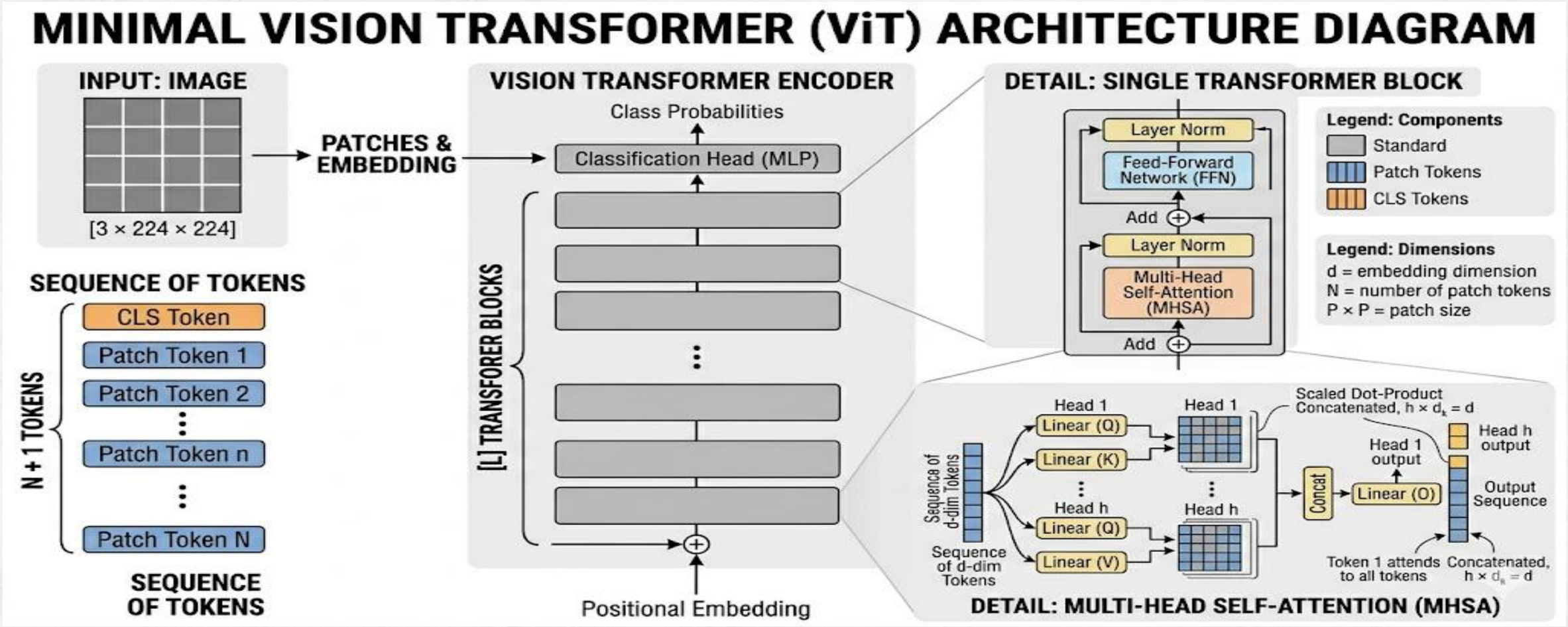
- **Spatial convolution:**
  - ➔ st= fs (Ws ∗ xt)
- **Temporal convolution:**
  - ➔ Ft = ft (Wt * Fst + bt)
- **Residual Learning:**
  - ➔ Yt = F (Fs1 + xt)
- **Final Spatial tOutput:**
  - ➔ Ys = ft (fs (Ws ∗ X)) + X

Feature Fusion



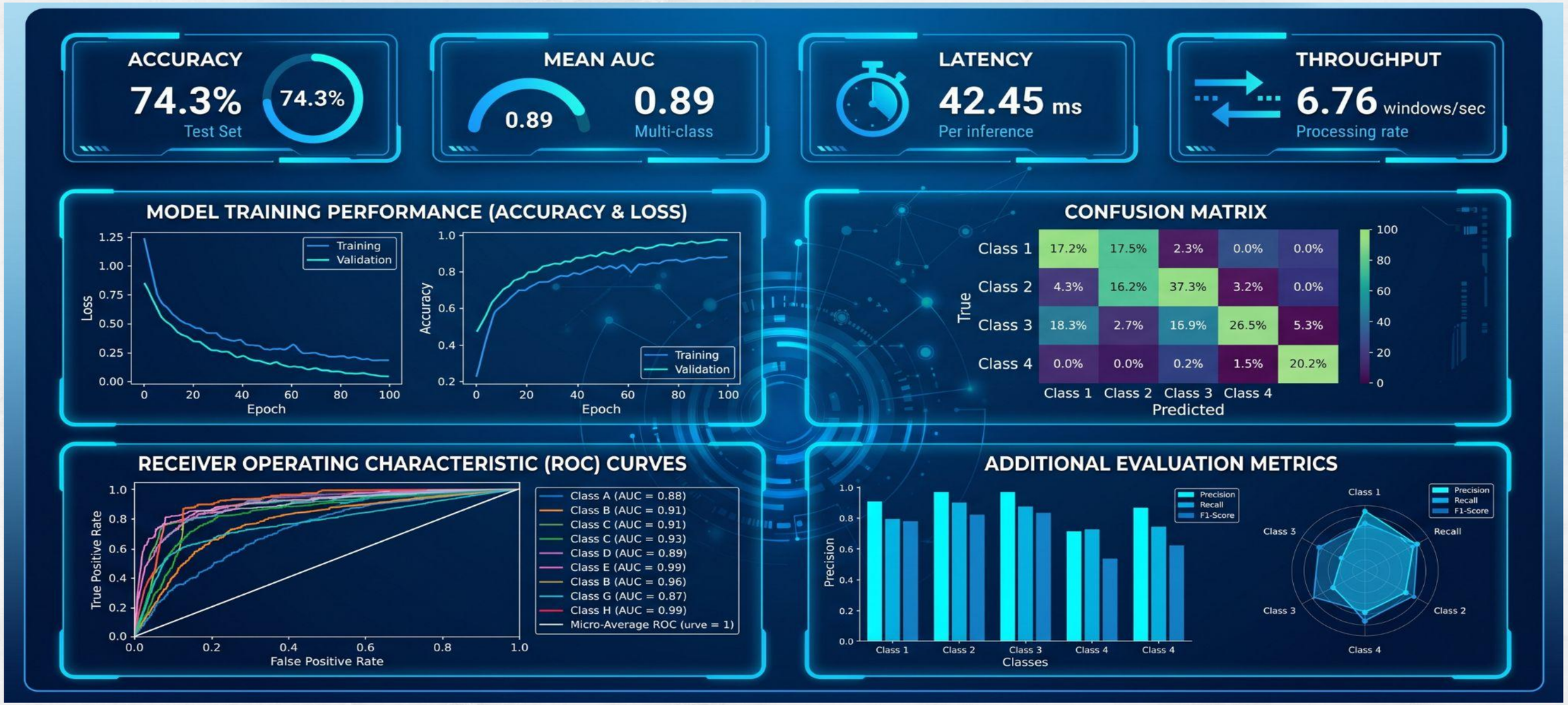
- **Equation (concatenation):**
  - ➔ F = [Fs;Ft]
- **Then projection:**
  - ➔ F′ = WF + b

Vision Transformer (ViT)



- Attention(Q,K,V)=softmax{Q*K(Transpose) / square root (d)}*V

Hybrid CNN-ViT Model Training process



Multi-Scenario Anomaly Detection (MSAD) Dataset

↑ Pros:

- Large-scale dataset for anomaly detection.
- Contains multiple surveillance scenarios and camera angles.
- Good for detecting whether something is normal or abnormal.

↓ Cons:

- No detailed activity classifications for example (theft, fire, fighting).
- Limited for training models that need specific alerts per activity type.

AIU ALAMEIN

UCF Crime Dataset (Chosen Dataset)

Pros:

- Contains large-scale real-world CCTV footage.
- Multiple activity classes: theft, fighting, fire, assault, etc.
- Suitable for training models for specific alerts.
- Well-documented and widely used in academic research.

Cons:

- Requires preprocessing to remove corrupted videos and balance classes.
- Requires feature extraction due to use of raw videos.

Why We Chose it:

- Real CCTV footage.
- Multiple activity types for precise detection.
- Large dataset for robust model training.



Chosen Dataset Description

- We are going to use the UCF Crime dataset







Dataset

❑ Dataset Images number by class **Training data**

Class	Number of images
Abuse	19.1k
Arrest	26.4k
Arson	24.4k
Assault	10.4k
Burglary	39.5k
Explosion	18.8k
ShopLifting	24.8k
Fighting	24.7
Normal	948k
RoadAccidents	23.5k
Robbery	41.5k
Shooting	7.1k
Stealing	44.8k
Vandalism	13.6k

- Total Training images in dataset **1,266,345 images/frame**

Dataset

❑ UCF Dataset Images number by class **Test data**

Class	Number of images
Abuse	297
Arrest	3365
Arson	2793
Assault	2657
Burglary	7657
Explosion	6510
ShopLifting	7623
Fighting	1231
Normal	65k
RoadAccidents	2663
Robbery	835
Shooting	7630
Stealing	1984
Vandalism	1111

- Total Training images in dataset 111,308 **images**

Th
eft Burglary, Robbery,
Shoplifting, Stealing

Data After Merging

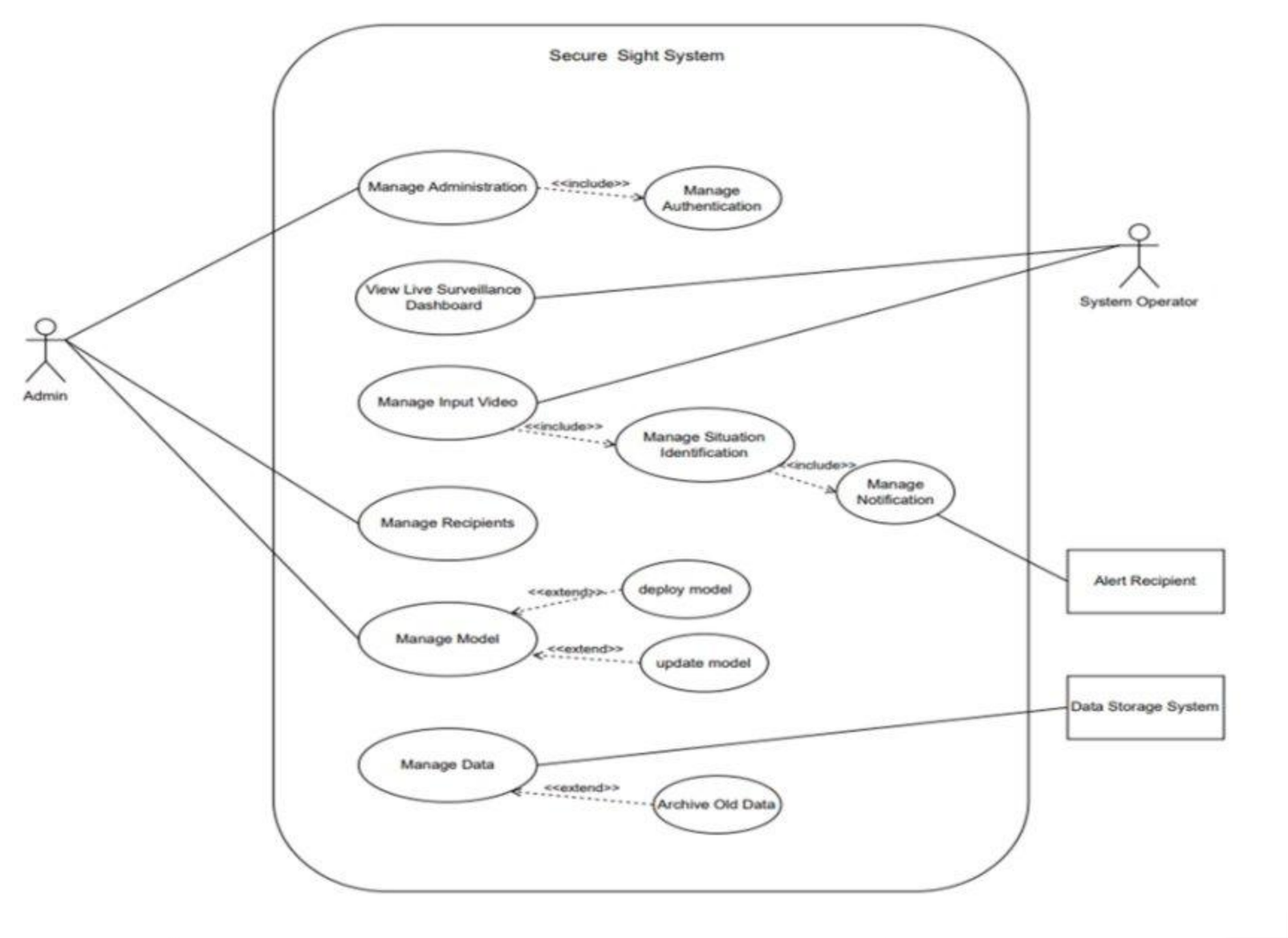
Merged Classes

- Theft:Burgalry,Robbery,Shoplifting,Stealing
- Violance:Abuse,Assault,Fighting
- Normal class downsampled by 40%

Class	Number of images
Theft	150.6k
Violence	54.2k
Arrest	26.4k
Arson	24.4k
Explosion	18.8k
RoadAccidents	23.5k
Shooting	7.1k
Vandalism	13.6k
Normal	Downsampled*

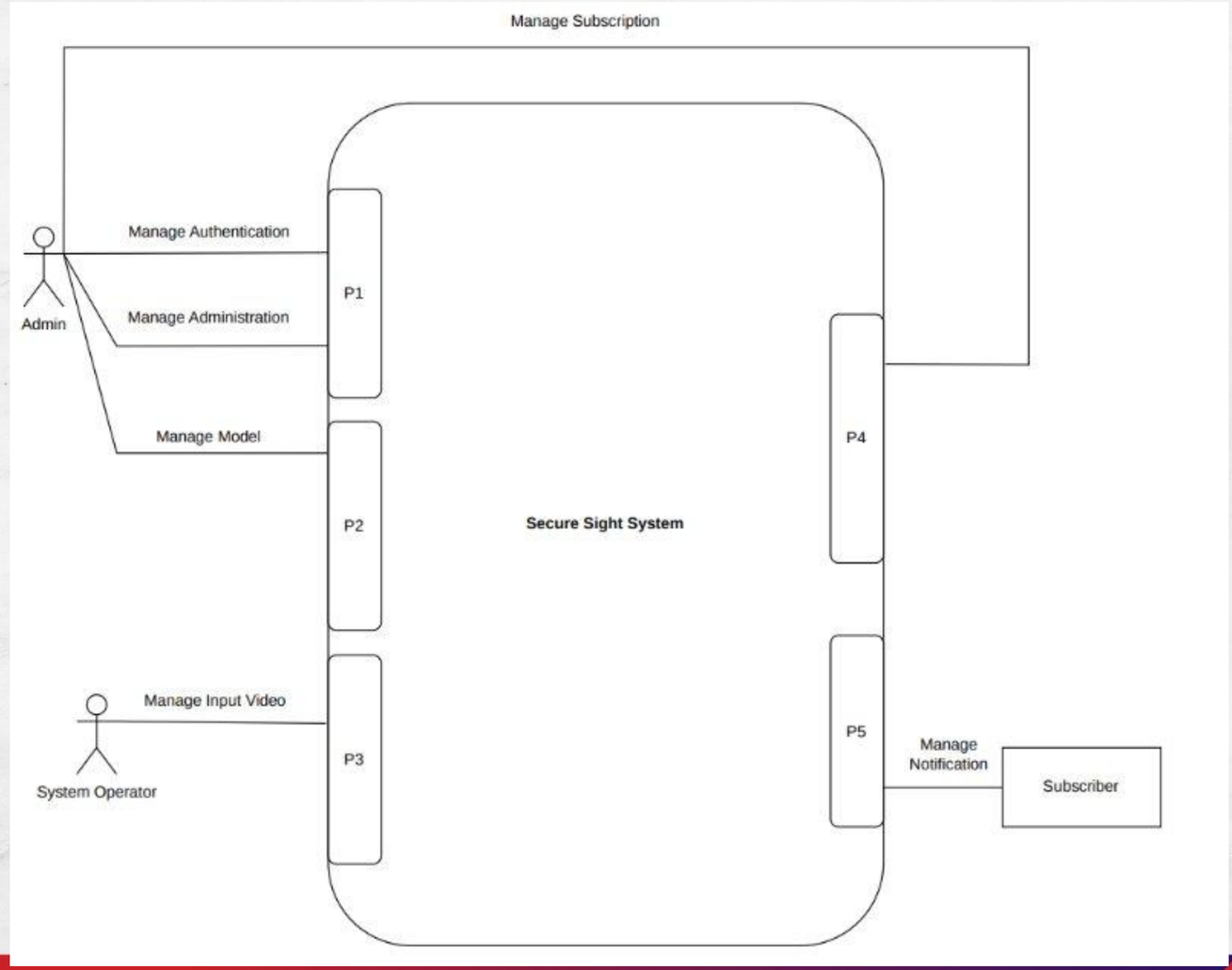
UML Diagrams

- Use Case
- Diagram



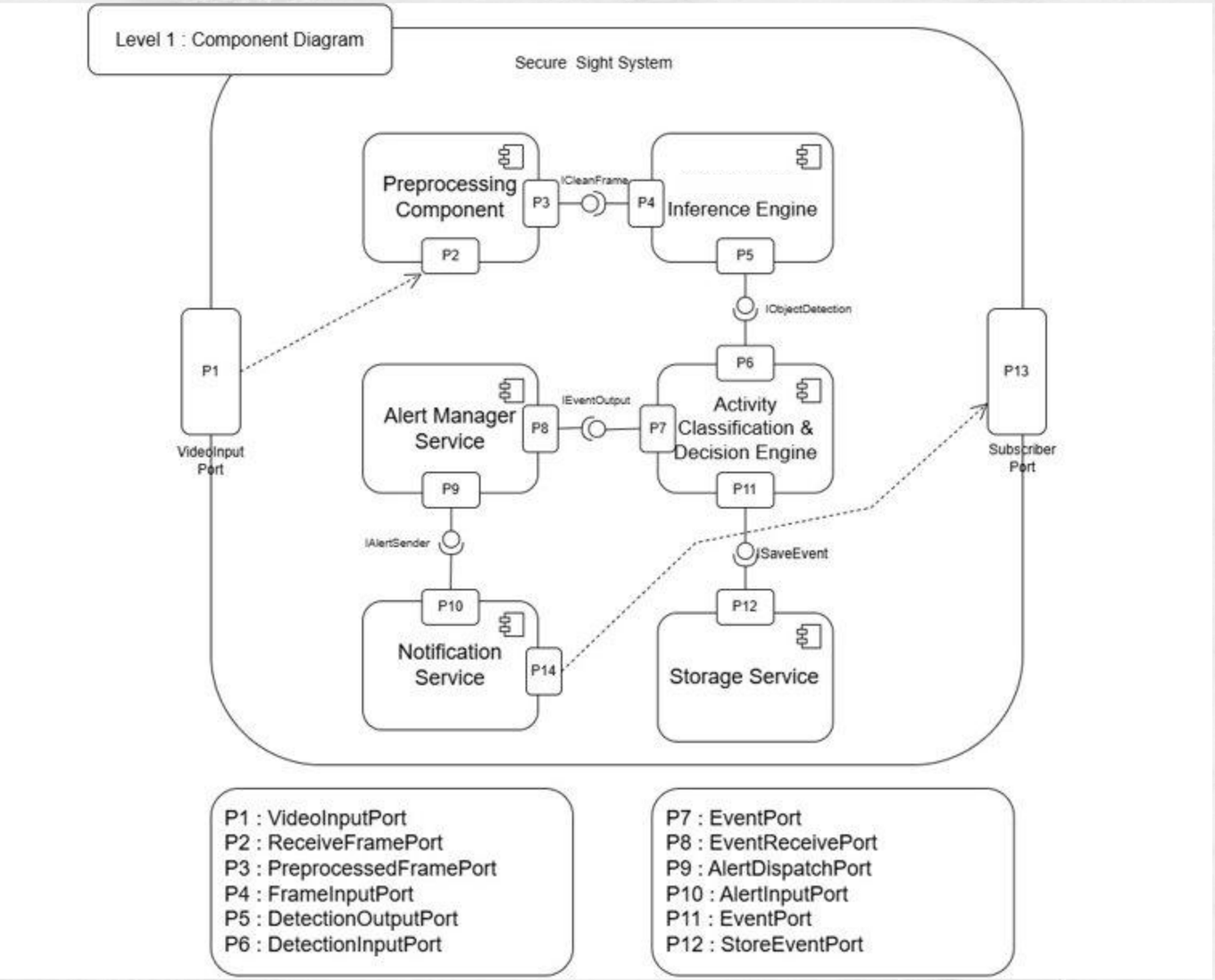
UML Diagrams

- Level 0 : Context
- Diagram

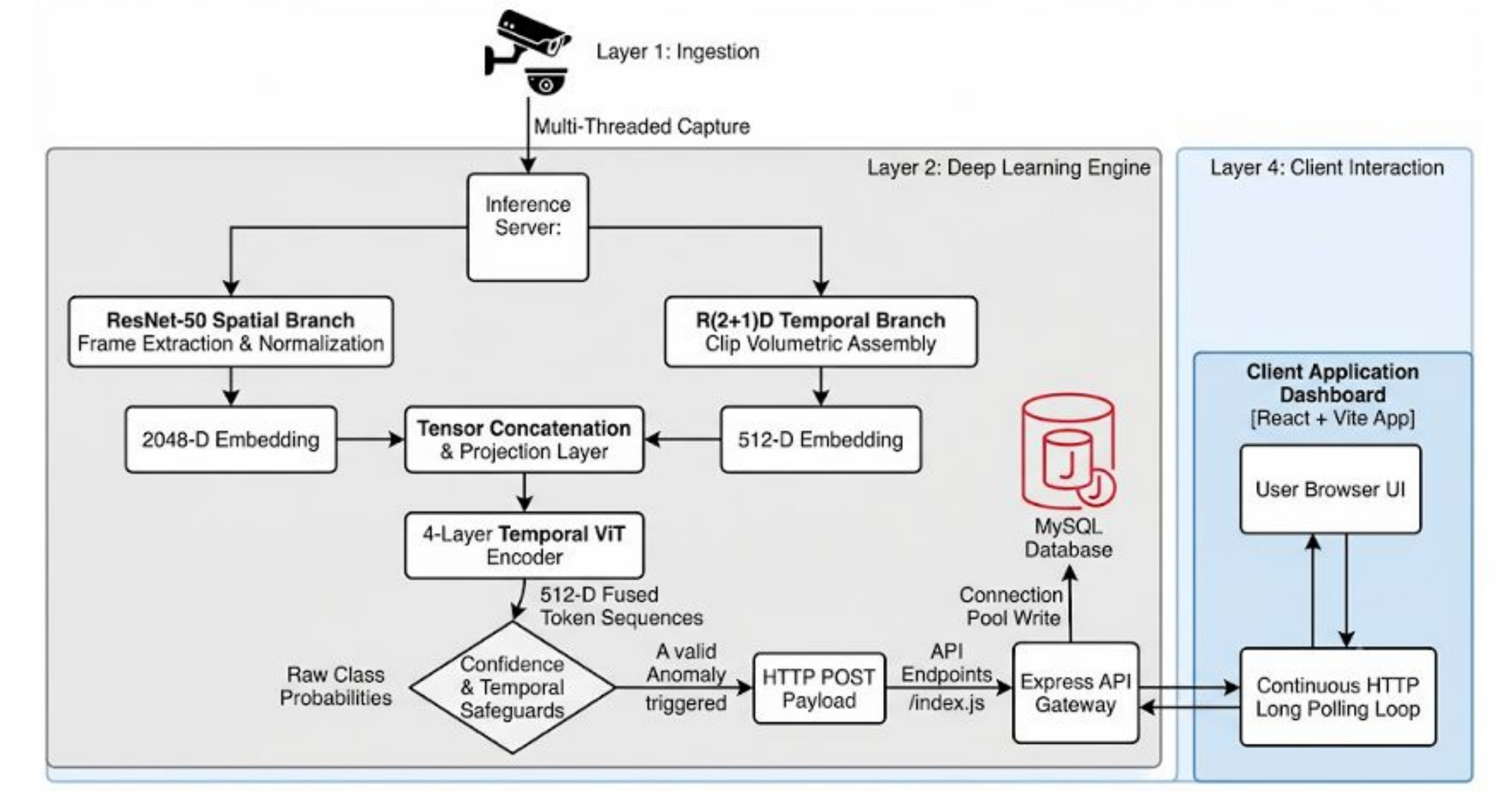


UML Diagrams

- Level 1 : Component
- Diagram



Implementation



Implementation

Web System Architecture

LIVE MODE

Integrated Data Flow & Backend API Gateway

Data Ingestion

- RTSP Stream Capture
- Multi-threaded Pipeline
- Frame Downsampling

SOURCE
IP Cameras (Edge)

Backend API Gateway

Routing Logic	Protocol
Async Dispatcher	**HTTP/WS**
Auth	**Load**
JWT Tokens	**Balanced**

Manages request queuing and response caching

System Integration

- Alert Dispatch Hub
- Service Orchestration
- Pub/Sub Messaging

TARGET
Responders / DB

Implementation

CHAPTER 5: IMPLEMENTATION DETAILS

AI Pipeline & Optimization

LATENCY 42ms

FRAMEWORK PyTorch

Dual-Branch Extraction

Spatial Branch

Static Appearance Features

`ResNet-50 (Pre-trained)`

Output: 2048-d Vector

Temporal Branch

Motion Dynamics (16-Frame Clips)

`R(2+1)D Factorization`

Output: 512-d Vector

CONCATENATION FUSION

2560-d Concat Vector

Linear Projection to 512-d

Vision Transformer

Sequence Modeling Layer

`4-Block Encoder`

Training Strategy

- **Clip-Level Supervision**
  Precise temporal annotations over weak labels.
- **Class Weighting**
  Inverse frequency sampling to mitigate normality bias.
- **PyTorch Lightning**
  Structured training loops & hardware scaling.

PERFORMANCE

Top-1 Acc: 74.3%

AUC: 0.89

Implementation

CHAPTER 5: IMPLEMENTATION DETAILS

Inference-Time Safeguards

DECISION ENGINE

Dual-layer verification mechanism to suppress false alarms while maintaining high detection accuracy.

Confidence Filter

THRESHOLD LEVEL

60%

Predictions below **60%** confidence are automatically discarded as background noise.

Temporal Validation

VERIFICATION WINDOW

5 sec

- Ensures anomaly persists consistently across consecutive frames.
- Filters short-term visual noise or momentary irregularities.
- Prevents false alerts triggered by sudden, non-threatening movements.

Input Score → Gate 1: Confidence → Gate 2: Duration → EMERGENCY ALERT

Implementation

Deployment & Robustness

Production Ready

Fault Tolerance

Designed to handle network failures gracefully with localized edge intelligence.

MECHANISM

Offline Buffering

Queues critical alerts locally during disconnects. **Zero data loss** upon network restoration.

Data Privacy Framework

PROCESSING MODE

Ephemeral

Volatile memory only. No disk persistence of raw streams.

RETENTION

Conditional

Clips stored **only if** anomaly alert is triggered.

Deployment

- Containerized Services
- GPU-Accelerated Inference
- Edge / Cloud Hybrid

Operator Dashboard

5.11

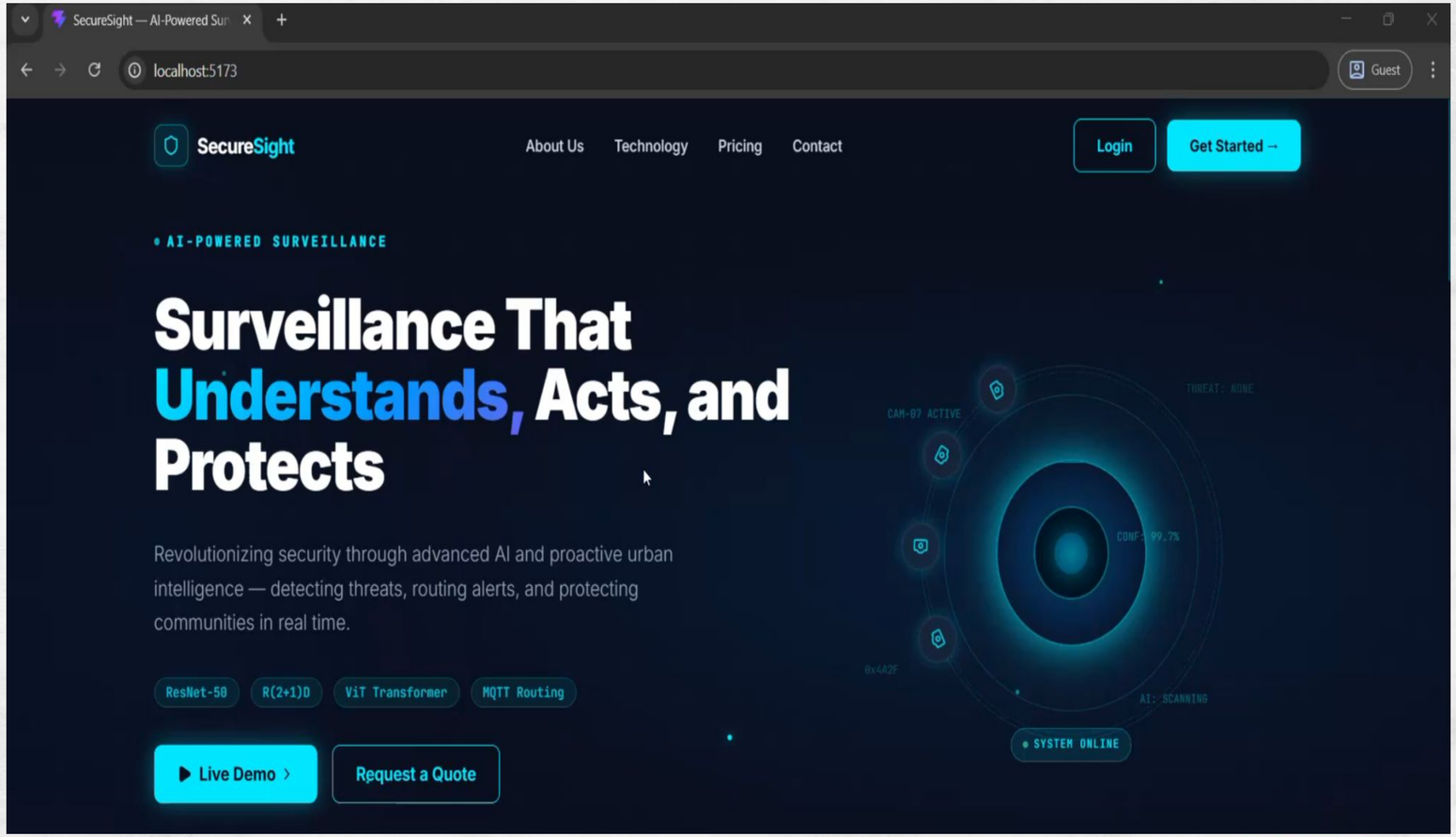
Live Feed Monitor

Real-time Stream Analysis

- Role-Differentiated Alerts
- Event Logging & History
- Threshold Configuration

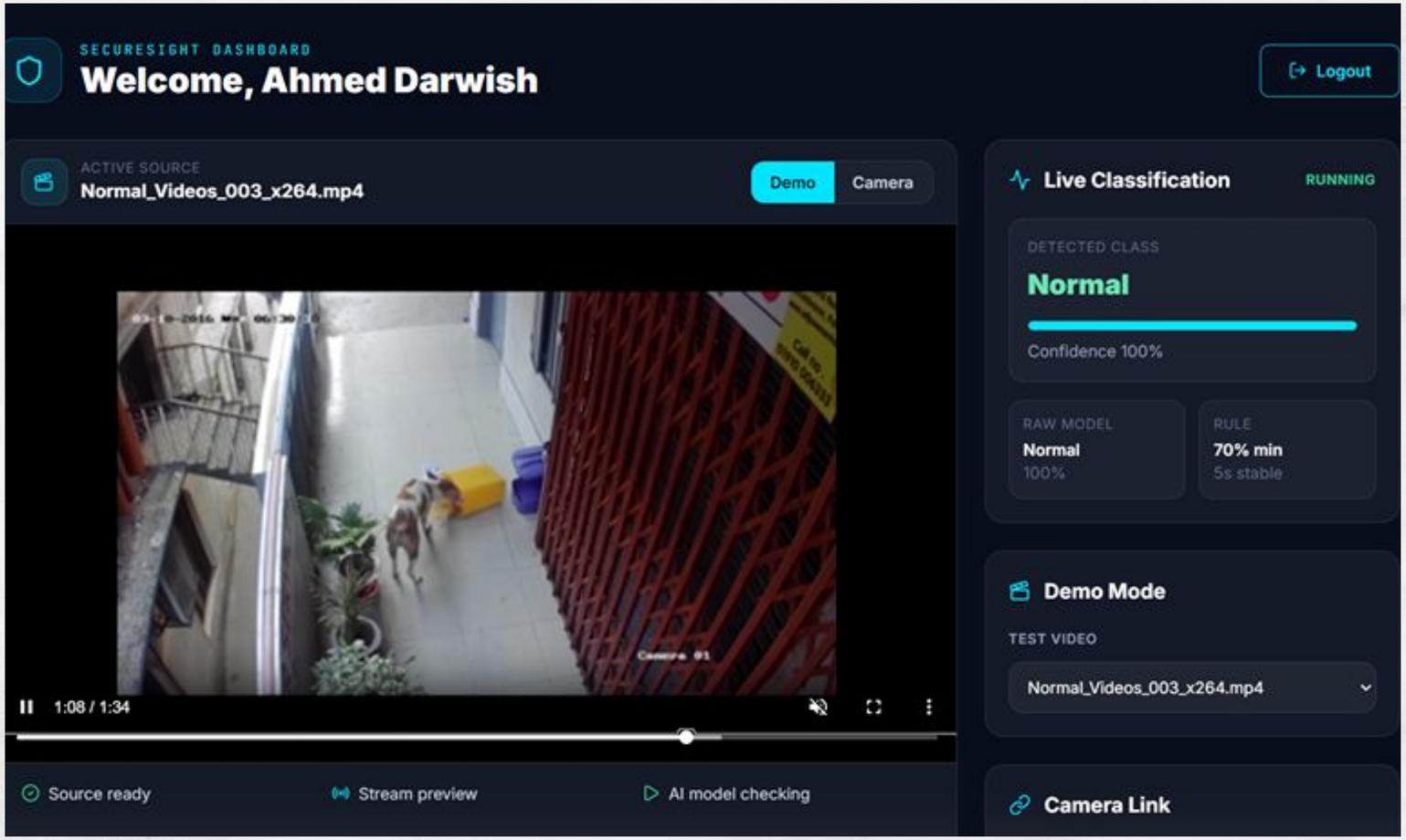
Web Interface

- End-to-end working system
- Real-time video analysis
- Multi-class anomaly detection
- Live alert visualization



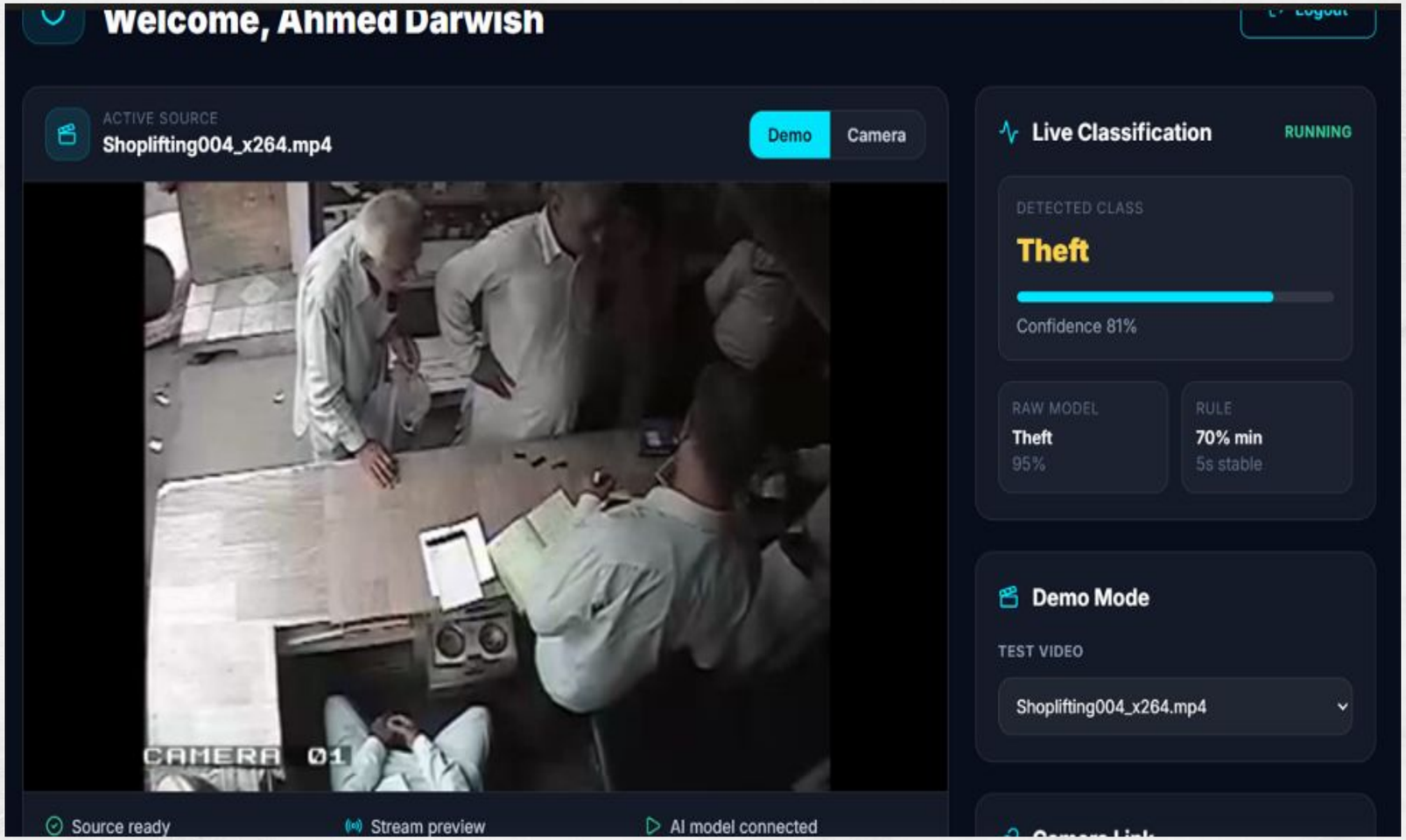
Web Interface

- Input: CCTV / video footage
- Model processes video stream
- Detects anomaly in real-time
- Displays confidence score
- Triggers alerts



Web Interface

- Input: CCTV / video footage
- Model processes video stream
- Detects anomaly in real-time
- Displays confidence score
- Triggers alerts



Results

74.3%
Top-1 Accuracy
UCF-Crime 14-class

0.89
Mean AUC
All anomaly classes

42ms
Processing latency
per window

Paper
Submitted
to journal

Dataset: UCF-Crime | 128h surveillance footage | 14 classes | 70/15/15 stratified split

Written Paper
Two-stream + ViT framework
paper accepted for journal submission

Comparison with Prior Work & Optimizer Study

TABLE 3 — Comparison vs. Prior Methods

Method	AUC
Sultani et al. 2018	75.41%
Zhong et al. 2019	82.12%
Tian et al. 2021	85.33%
SecureSight (Ours)	**89.0%**

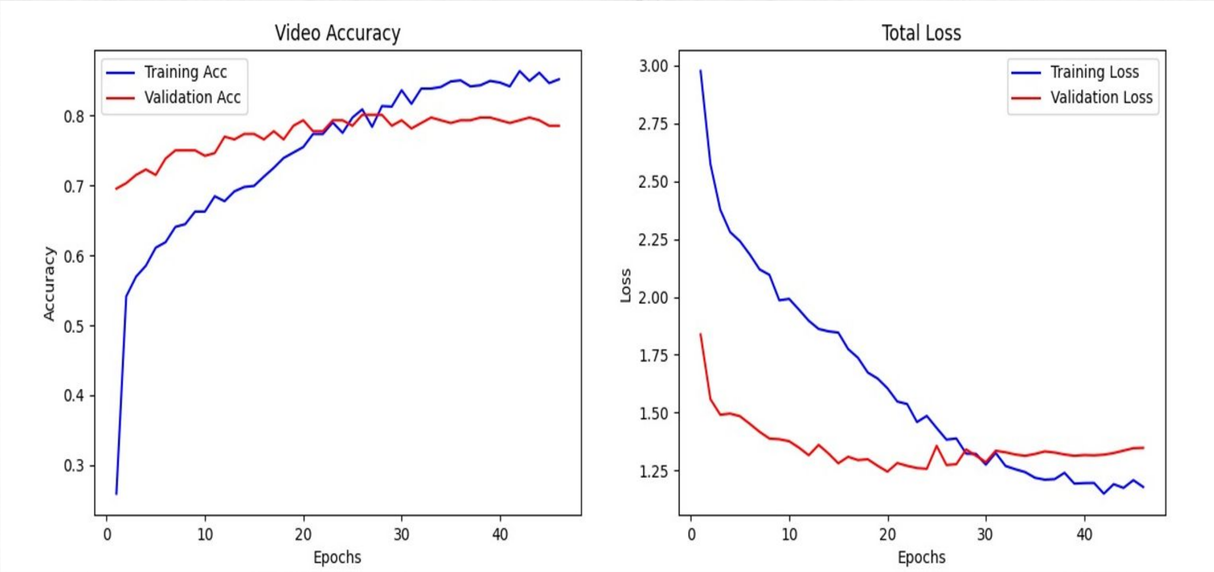
**Supervised clip-level annotations vs. weakly supervised prior work*

TABLE 4 — Meta-Heuristic Optimizer Results

Optimizer	Val Acc
Hand-tuned (baseline)	**74.3%**
GWO	70.8%
PSO	70.2%
HHO	69.9%
SMO	69.4%
SFLA	68.7%

All 5 optimizers failed to beat hand-tuned config
Negative result — reported transparently

Training Convergence

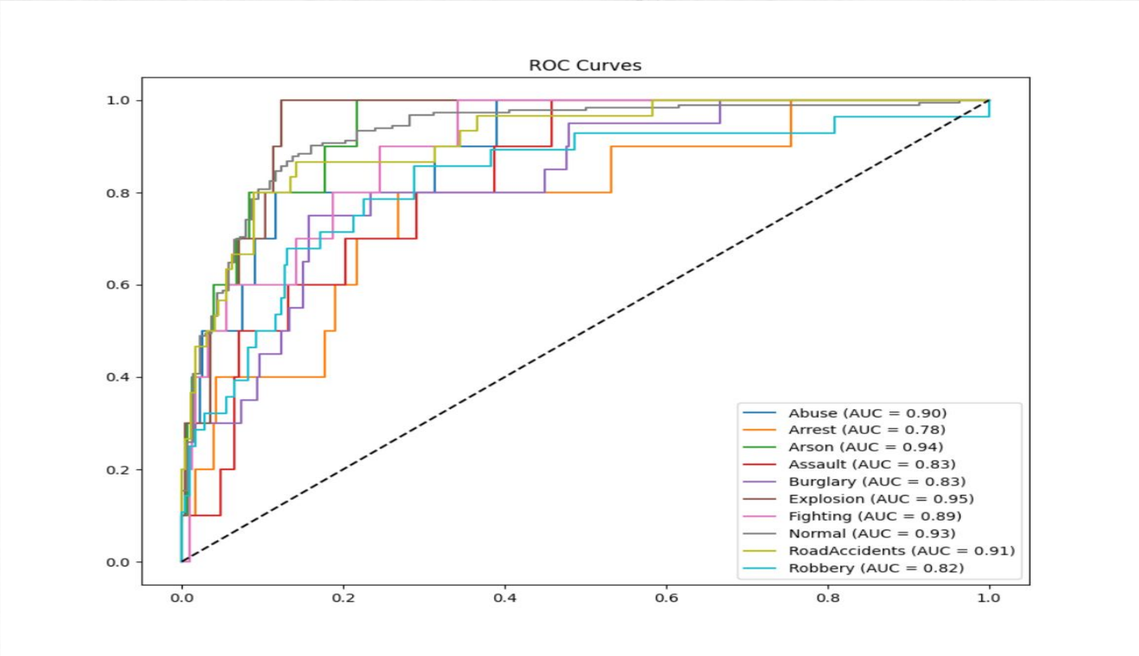


Training & Validation Curves — 60 epochs

TABLE 2 — Ablation: Component Contributions

Model Configuration	Val Acc (%)
ViT only — no annotations	57.2
ViT only — with annotations	70.3
ResNet-50 spatial only	63.5
R(2+1)D temporal only	67.1
Two-stream fusion (no ViT)	72.8
Two-stream + ViT (OURS)	**74.3**

Per-Class AUC — ROC Analysis



ROC curves — UCF-Crime test set

TABLE 1 — Per-Class AUC Values

Class	AUC
Abuse	0.90
Arrest	0.78
Arson	0.94
Assault	0.93
Burglary	0.83
Explosion	0.95
Fighting	0.89
Normal	0.93
RoadAccidents	0.91
Robbery	0.82
Mean AUC	**0.89**

Lowest: Arrest (0.78) & Robbery (0.82) — visual similarity to normal activity

paper acceptance letter

Dear Mohamed A hassona,

We are pleased to inform you that your paper A Real-Time Intelligent Surveillance System for Video Anomaly Detection Using Two-Stream Deep Learning Fusion and Vision Transformers has been accepted for online presentation at AiA26 International Conference.
Congratulations!

To proceed, we kindly ask you to:

- Submit the camera-ready version of your paper, incorporating the reviewers' comments and following the conference formatting guidelines.

- Prepare your online presentation, ensuring clear audio/video quality and adherence to the conference instructions (duration : 10 min, format : MP4, email to : aia-doctoralday@uae.ac.ma).

- It is mandatory to be present online during your session, especially for the live Q&A discussion following your presentation, to interact with the audience and answer questions.

Please ensure that the camera-ready version is submitted by April 1st, 2026.

Additional information regarding the presentation schedule and technical guidelines will be shared soon.

We look forward to your participation and your online presentation.

Kind regards,
AiA26 Steering Committee.

Please do not reply to this email as it was generated from an email account that is not monitored.

To stop receiving conference emails, you can check the 'Do not send me conference email' box from your User Profile.

Microsoft respects your privacy. To learn more, please read our Privacy Statement.

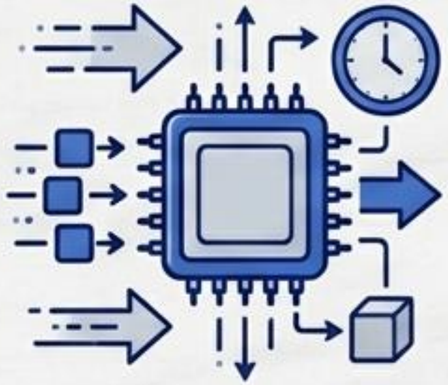
Microsoft Corporation
One Microsoft Way
Redmond, WA 98052

Challanges



Data Quality

- Accuracy & Completeness
- Consistency
- Validation
- Standardization



Real-Time Processing

- Low-Latency Ingest
- In-Stream Analytics
- Instant Decisioning

False Positives/Negatives

- Minimize Incorrect Alerts
- Reduce Missed Threats
- Optimize Thresholds





System Integration

- Seamless API Connections
- Cross-Platform Sync
- Data Silo Elimination

Solutions

DATA QUALITY SOLUTIONS



PREPROCESSING PIPELINE

REAL-TIME PROCESSING SOLUTIONS



Latency Drop: 150ms → 28ms

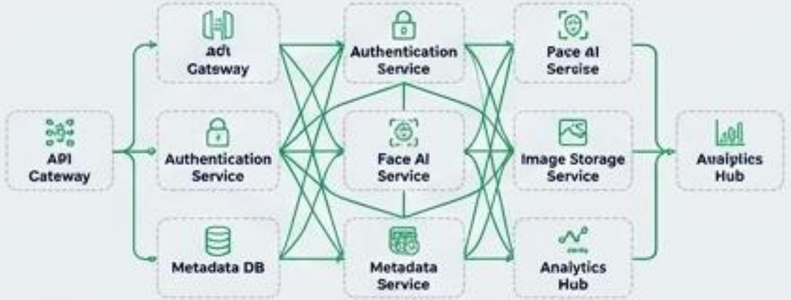
PERFORMANCE OPTIMIZATION

FALSE ALERTS REDUCTION



THRESHOLD STRATEGY

SYSTEM DEPLOYMENT SOLUTIONS



MICROSERVICES ARCHITECTURE

Project Timeline

Phase 1: Project Initiation (Research & Scope)

Phase 2: Planning & Design (Strategy & Outlines)

Phase 3: Execution (Development & Implementation)

Phase 4: Integration (System Assembly & Testing)

Phase 5: Final Review (Validation & Quality Control)

Phase 6: Launch (Deployment & Delivery)

Team Contributions

MOHAMED AHMED
Software Engineer & Team Lead

Coordinated overall architecture, managed system integration.

OMNIA ADEL
AI Engineer

Co-developed spatial apearance pipelines, optimized pre-trained ResNet-50 backbone.

TASNEEM IBRAHIM
AI Engineer

Assisted with temporal data training, structured dataset preprocessing pipelines.

ABDULLAH ABDALGAWAD
AI Engineering Lead

Designed feature extraction architecture, engineered R(2+1)D and ResNet-50 temporal pipelines, led deep learning model training.

KHALID ABDELSALAM
Software Engineer

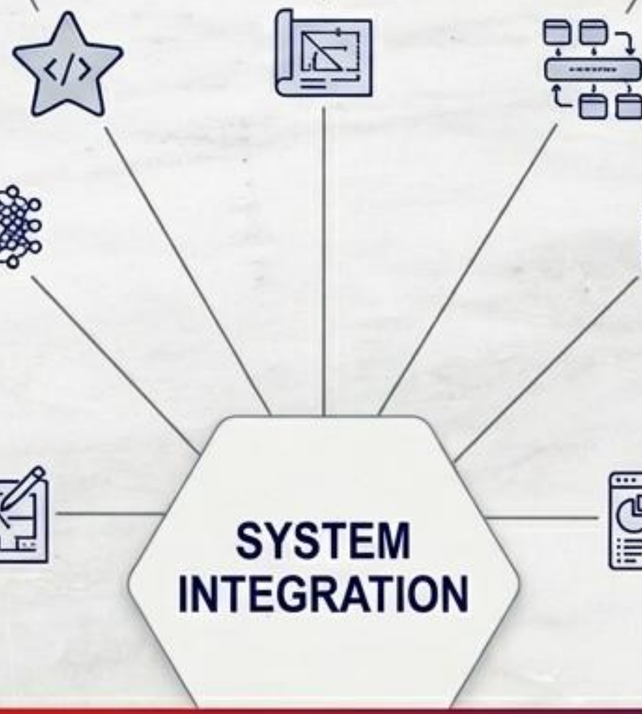
Implemented local data optimized management rules, offline edge buffering mechanisms.

AHMED MOSTAFA
Software Engineer

Co-developed application, englneered system data pathways, designed system architecture.

MARIAM ELRAFEI
Software Engineer

Designed live operator dashboards, handled system latency profiling.



Limitations



DETECTION ACCURACY

Depends on camera quality & environmental conditions



CROWD PERFORMANCE

May decrease in highly crowded scenes



HARDWARE RESOURCES

Real-time processing requires adequate resources



DATA REQUIREMENTS

Some rare object classes may require additional training data

AIU ALAMEIN INTERDISCIPLINARY UNIVERSITY جامعة الألفين الدولية

Future Work

Support for Multiple Camera Streams

Multi-stream support: Up to 50 inputs

Cloud-Based Deployment

Scalable Cloud Architecture

Mobile Application Integration

Native iOS & Android apps

Advanced Predictive Analytics

Pre-emptive Threat Scoring

Continuous Learning & Model Updates

Automated On-Device Training

Enhanced Threat Detection Capabilities

Multi-modal sound & motion logic

Conclusion



Computer Vision + Deep Learning = Smarter, Reliable Surveillance

References

- UCF-Crime Dataset – Sultani et al., CVPR 2018
- UCA Annotation Dataset – Yuan et al., arXiv 2023
- ResNet-50 – He et al., CVPR 2016
- I3D – Carreira & Zisserman, CVPR 2017
- R(2+1)D – Tran et al., CVPR 2018
- Vision Transformer (ViT) – Dosovitskiy et al., arXiv 2020
- Graph Convolutional Label Noise Cleaner – Zhong et al., CVPR 2019
- RTFM Method – Tian et al., ICCV 2021